Adjustable Iris Shutter for 10 m Air Pistol Shooting Glasses: Design Report

Status: CAD prototype only (Onshape). Nothing has been printed, built or tested physically.
Report date: 2026-11-07 **Purpose of this document:** record every dimension, component and design decision, and how reliable each one is, so the design can be checked against the current ISSF rulebook before production or tournament use.

> **Legality:** this report makes **no claim** that the design is legal or illegal. No rulebook research went into it. The only rules-related statement ever retrieved came from a secondary source (Wikipedia) and says irises on glasses are allowed if not mounted on the pistol. The actual rulebook text and any dimensional limits were **never read**. Verify against the current ISSF rulebook, and consider asking an equipment-control official before production.

Provenance tags

Tag	Meaning
[CAD]	Measured or confirmed by the author in Onshape
[USER]	Stated by the author in conversation, not independently checked
[DERIVED]	Calculated from other values
[CHOSEN]	Starting parameter picked during design, with no external source
[PLACEHOLDER]	Estimate that needs real-world verification
[UNVERIFIED]	Not confirmed

All values below are in **millimetres**. Onshape dialogs in this project often displayed centimetres, which caused several unit mix-ups during design. Check document units before reusing any number.

1. Device summary

- Manually adjustable iris diaphragm, **10 blades**, intended to sit in front of the lens of shooting glasses (not on the pistol).
- Purely mechanical **pin-and-slot linkage** (no gears or racks). Blades pivot on fixed pins. Guide pins on a rotating ring slide in a slot in each blade.
- No lens, optics, electronics or coatings.
- Stepless adjustment, no detents, friction only. No locking mechanism was designed.
- Aperture is a **10-sided polygon** (straight blade edges), not a circle.
- Adjusted by sliding a tab through an arc slot in the casing.
- Production method undecided. Blades were conceived as laser-cut or etched sheet. Rings could be printed or machined.

2. Aperture performance

Item	Design	Measured
Fully open	12.0 mm Ø	about 12.1 mm Ø (6.04 mm radius) [CAD]
Fully closed	1.0 mm Ø	about 1.2 mm Ø (0.58 mm radius) [CAD]
Actuator ring rotation, open to closed	about 28.4°	[DERIVED]
Maximum blade rotation	54.8°	[DERIVED]

Aperture readings are from the CAD Measure tool. **Physical confirmation is deferred** by the author.

3. Overall size

Item	Value	Status
Casing outer diameter	47	[CHOSEN]. Enlarged from the original 37 mm concept so the blades fit.
Cavity inner diameter	43	[CHOSEN]
Rim wall thickness	2.0	[DERIVED]
Rim height	1.9	[CHOSEN]
Front ring base plate thickness	1.0	[CHOSEN]
Casing offset allowance	about 0.02	[USER], added to absorb overlap once physically built
Total thickness	not recorded	**[UNVERIFIED]**. Measure in CAD, including the rear ring and lug.

The 47 mm casing is about 10 mm wider than the 37 mm lens holder. Its effect on the shooter's peripheral view has not been assessed.

4. Components

1. **Front ring:** fixed. Carries 10 pivot pins, the rim, the actuator-tab arc slot and the cover-pin holes.
2. **Actuator ring:** rotates. Carries 10 guide pins, an "L-R" marking and the actuator tab.
3. **10 blades:** flat sheet with a pivot hole and a guide slot each.
4. **Cover plate:** closes the cavity.
5. **Actuator tab:** protrudes through the rim slot.
6. **4 friction-fit wire pins:** retain the cover plate.

7. **Rear features:** raised ring and mounting lug (both placeholders, see section 8).

5. Blade (x10)

Feature	Value	Status
Outline before trim	13 x 14.6 rectangle, x from -7 to 6, y from -2 to 12.6, origin at pivot hole	[CHOSEN]
Corner trim	line from (6, 8.94) to (3.8, 12.6)	Applied [USER]. Final dimensions not recorded.
Thickness	0.1	[CHOSEN]
Pivot hole	Ø0.9 at origin	[CHOSEN]
Slot width	0.9	[CHOSEN]
Slot end-circle centres	(-1.72, 4.10) and (4.43, 9.27)	[DERIVED]
Slot length, centre to centre	8.03 design, 8.047 measured	[CAD]
Slot direction	40° from blade x-axis	[CHOSEN]
Working edge	straight, 7.0 from pivot centre	[DERIVED]

6. Pins, rings and linkage

Feature	Design	Measured	Status
Pivot-pin circle radius	13.0	12.985	[CAD]
Guide-pin circle radius	12.0	12.010	[CAD]
Slot start to pivot centre	4.45	4.446	[CAD]
Pin count	10 pivot, 10 guide, 36° apart	as modelled	[CHOSEN]
Pin diameter	0.8	not re-measured	[CHOSEN]
Pin height	0.30	not re-measured	[CHOSEN]
Actuator ring	Ø32 outer, Ø13 bore, 1.0 thick	as modelled	[CHOSEN]
Ring-to-ring gap	0.35	not re-measured	[CHOSEN]
Blade layers	two layers (alternating)	not verified in the model	[CHOSEN]

Linkage mates in Onshape: Revolute (ring to ring), Cylindrical (blade to pivot pin), Pin Slot (guide pin to blade slot, translation limit 0 to 8.03).

7. Cover plate, tab and cover retention

Feature	Value	Status
Cover plate	Ø47 x 1.0	[CHOSEN]. Built per author; exact values unconfirmed.
Register step	Ø42.8 x 0.5	[CHOSEN]
Central bore	Ø13	[CHOSEN]
Actuator tab	about 2 wide, radial from Ø32 edge to about Ø44	[CHOSEN]. Current tab described as a stopgap.
Rim arc slot	rebuilt so the tab base does not touch the rim wall	[USER]. **Dimensions not provided.**
Cover-pin holes	4 x Ø1.0 on a 22.53 radius, 90° apart, through the full casing	[USER]
Hole wall margins	0.47 to 0.53	[DERIVED] (rim spans radius 21.5 to 23.5)
Holes vs tab slot	clear of the arc slot	[USER]
Cover retention	4 friction-fit wire pins, not glued	Design decision
Estimated pin length	about 3.9 (cover 1.0 + rim 1.9 + base plate 1.0)	[DERIVED], **[UNVERIFIED]** in the model

The casing is openable by pushing the pins out. Pin diameter, retention strength and wear are unverified. A permanent or screw-based method is still undecided. Standard 1.75 mm filament does not fit these holes.

8. Attachment to the glasses (incomplete)

Feature	Value	Status
Target interface	37 mm lens holder	The original concept image labelled it "threaded". A manufacturer listing (Gehmann, Knobloch accessories) describes 37 mm accessories as clip-on or push-on, so the thread claim appears wrong.
Rear feature	Ø37 / Ø34 ring, now extruded outward from the back face	[USER]. Height not provided. No verified fit to any real holder. [PLACEHOLDER]
Rear mounting lug	10 x 6 x 3 tab with a Ø2.7 hole	[PLACEHOLDER]
Frame clip	not designed	Needs caliper measurements of the real clip or frame

9. Rule-relevant parameters to check

Categories I would expect a rulebook check to touch. None has been researched.

Item to check	Rule reference	Verified
Iris permitted on glasses, and conditions		
Limits on iris diameter or aperture range		
Limits on outside dimensions (47 mm casing)		
Limits on thickness or distance in front of the eye		
Whether the casing counts as a side blinder or eyeshield		
Materials, finish, reflectivity, colour		
Aperture shape (10-sided polygon)		
Marking or equipment-control requirements		
In-match adjustment allowed; locking required or forbidden		
Any attachment to the pistol (none designed)		
Blinder over the non-shooting eye: 30 mm limit (author-stated; applies to the blinder, not the iris casing). Not read in the rulebook.		
Whether the 47 mm casing could be treated as a blinder or block the non-shooting eye		

10. Open items

Item	Status
Interference check	**Not run.** Author judged it unnecessary from visual checks; a 0.02 mm offset is used instead. Overlaps between stacked layers cannot be seen from the top view.
Total thickness	Not recorded
Rim arc-slot dimensions	Not provided
Raised rear ring height	Not provided
Cover-pin diameter and fit	Untested
Frame clip and lug	Not designed or measured
Printability	Deferred. 0.1 mm blades, 0.8 mm pins and 0.05 mm clearances are unlikely to print on a typical hobby printer.
Physical aperture test	Deferred
Lens-holder fit	Unverified against real hardware
Non-shooting-eye blind spot	To be checked on the physical prototype (author's plan). Check with both eyes open, iris fully open, at the real stance head position.
Iris size limit	None found. The 2026 Pistol Rules chapter (Ch. 8) has no iris or blinder dimension. The General Technical Rules were not retrieved.

11. Design history notes

- The reference concept image was AI-generated and its own note says it is not for fabrication.
- Early blade geometry overhung the Ø37 rings by several mm. Fixed by enlarging the casing to Ø47 and trimming one blade corner.
- The aperture initially stalled at about 10 mm open and 2 mm closed. Cause found: blades had settled in a flipped orientation, plus the working edge sat 1 mm too close to the pivot. Both corrected.
- Several early over-defined mate errors were cleared by switching the blade-to-pivot mate from Revolute to Cylindrical.

12. Sources

- Onshape CAD model and measurements by the author.
- Gehmann, Knobloch shooting frames and accessories listings (37 mm clip-on fit).
- Everything else is derived from calculation or stated by the author.